

定积分证明方法小结

定积分等式证明：应用介值定理、微分和积分中值定理

例 设 $f(x)$ 在 $[a, b]$ 上连续, 且 $f(x) > 0$, 证明方程 $\int_a^x f(t)dt + \int_b^x \frac{1}{f(t)}dt = 0$, 在

(a, b) 内有且仅有一个实根。

$$F'(x) = f(x) + \frac{1}{f(x)} \geq 2$$

$$F(a) = \int_b^a \frac{1}{f(t)}dt < 0$$

$$F(b) = \int_a^b f(t)dt > 0$$

证 存在性: 设 $F(x) = \int_a^x f(t)dt + \int_b^x \frac{1}{f(t)}dt$, 由题设知 $F(x)$ 在 $[a, b]$ 上连续, 且

$$F(b) = \int_a^b f(t)dt > 0; \quad F(a) = \int_b^a \frac{1}{f(t)}dt < 0$$

由零点定理必有 $F(\xi) = 0, \xi \in (a, b)$

唯一性: $F'(x) = f(t) + \frac{1}{f(t)} \geq 2$, 故 $F(x)$ 在 (a, b) 内单调增加, 零点 ξ 唯一

例 设 $f(x)$, $g(x)$ 在 $[a, b]$ 上连续, 证明至少 $\xi \in (a, b)$, 使得

$$(1) \quad f(\xi) \int_{\xi}^b g(x) dx = g(\xi) \int_a^{\xi} f(x) dx; \quad (2) \quad \frac{\int_a^b f(x) dx}{\int_a^b g(x) dx} = \frac{f(\xi)}{g(\xi)}, \quad g(x) \neq 0$$

Handwritten notes: $f(\xi) \int_a^b g(x) dx - g(\xi) \int_a^{\xi} f(x) dx$, $F(x) = \int_a^x f(t) dt$, $G(x) = \int_a^x g(t) dt$, $F(b) - F(a) = \int_a^b f(x) dx$, $G(b) - G(a) = \int_a^b g(x) dx$, $F'(x) = f(x)$, $G'(x) = g(x)$, $F(a) = F(b) = 0$, $G(a) = 0$, $\Rightarrow \frac{F(b) - F(a)}{G(b) - G(a)} = \frac{F'(\xi)}{G'(\xi)}$

证 (1) 由于 $f(\xi) \int_{\xi}^b g(x) dx - g(\xi) \int_a^{\xi} f(x) dx = \left[\int_a^x f(t) dt \int_x^b g(t) dt \right]_{x=\xi}'$

设 $F(x) = \int_a^x f(t) dt \int_x^b g(t) dt$, 显然 $F(x)$ 在 $[a, b]$ 上连续, 在 (a, b) 内可导, 且

$F(a) = F(b) = 0$, 由罗尔定理至少 $\xi \in (a, b)$, 使得 $F'(\xi) = 0$, 即

$$f(\xi) \int_{\xi}^b g(x) dx = g(\xi) \int_a^{\xi} f(x) dx$$

(2) 设 $F(x) = \int_a^x f(t) dt$, $G(x) = \int_a^x g(t) dt$, 显然 $F(x)$, $G(x)$ 在 $[a, b]$ 上满足柯

西条件, 则

$$\frac{\int_a^b f(x) dx}{\int_a^b g(x) dx} = \frac{F(b) - F(a)}{G(b) - G(a)} = \frac{F'(\xi)}{G'(\xi)} = \frac{f(\xi)}{g(\xi)}, \quad \xi \in (a, b)$$

例 设 $f(x)$ 在 $[a, b]$ 上连续, $\int_a^b f(x)dx = 0$, $\int_a^b xf(x)dx = 0$, 求证: 在 (a, b) 内

至少存在两点 $\xi_1 \neq \xi_2$, 使得 $f(\xi_1) = f(\xi_2) = 0$

证法 1 令 $F(x) = \int_a^x f(t)dt$, 则 $F'(x) = f(x)$, 且 $F(a) = F(b) = 0$, 又

$$\int_a^b xf(x)dx = \int_a^b x dF(x) = xF(x) \Big|_a^b - \int_a^b F(x)dx = -\int_a^b F(x)dx = 0$$

由积分中值定理有

$$\int_a^b F(x)dx = F(\xi)(b-a) = 0 \Rightarrow F(\xi) = 0, \xi \in (a, b)$$

于是 $F(a) = F(\xi) = F(b) = 0$, 对 $F(x)$ 在 $[a, \xi]$, $[\xi, b]$ 上分别应用罗尔定理得

$$F'(\xi_1) = f(\xi_1) = 0, \xi_1 \in (a, \xi); F'(\xi_2) = f(\xi_2) = 0, \xi_2 \in (\xi, b)$$

72. 设 $f(x)$ 在 $[a, b]$ 上连续, 且 $f(a) \leq f(x) \leq f(b)$, A 是 $f(x)$ 在 $[a, b]$ 上的平均值。证

明: 存在 $\xi \in [a, b]$, 使得 $A = \frac{f(a)(\xi - a) + f(b)(b - \xi)}{b - a}$

证 1: 即证 $A(b - a) = \int_a^b f(x) dx = f(a)(\xi - a) + f(b)(b - \xi)$

Handwritten notes:
 $A = \frac{\int_a^b f(x) dx}{b - a}$
 $\Rightarrow \int_a^b f(x) dx = f(a)(\xi - a) + f(b)(b - \xi)$
 $\Rightarrow (b - a)A = (f(a) - f(b))\xi + f(b)b - f(a)a$
 $\leq (b - a)f(b)$

设 $F(x) = f(a)(x - a) + f(b)(b - x)$, 显然 $F(x)$ 在 $[a, b]$ 上连续, 由于

$f(a) \leq f(x) \leq f(b)$, 所以

$$F(b) = f(a)(b - a) \leq \int_a^b f(x) dx \leq \int_a^b f(b) dx = f(b)(b - a) = F(a)$$

于是由介值定理, 存在 $\xi \in [a, b]$, 使 $\int_a^b f(x) dx = F(\xi) = f(a)(\xi - a) + f(b)(b - \xi)$

定积分不等式的证明

常用定理：定积分的比较定理，估值定理，函数单调性判别法，微分与积分中值定理，泰勒公式；

常用不等式： $a^2 + b^2 \geq 2ab$, $a + \frac{1}{a} \geq 2$ ($a > 0$)

柯西不等式： $\left[\int_a^b f(x)g(x) dx \right]^2 \leq \int_a^b f^2(x) dx \int_a^b g^2(x) dx$

常用等式： $b - a = \int_a^b dx$, $f(x) = \left(\int_a^x f(t) dt \right)'$, $\ln \frac{b}{a} = \int_a^b \frac{1}{x} dx$

定积分不等式的证明

题型一：利用换元法、分部法或周期函数的定积分性质直接求证；

4. 若 $f(x)$ 在 $[0,1]$ 上连续且单调下降, 证明, 对于任意的 $a \in (0,1)$, 都有

$$\int_0^a f(x) dx \geq a \int_0^1 f(x) dx.$$

$$\text{令 } F(x) = \int_0^x f(t) dt - x \int_0^1 f(t) dt$$

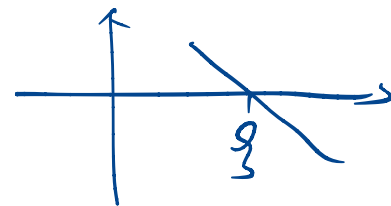
$$F(0) = 0 \quad F(1) = 0$$

$$F'(x) = f(x) - \int_0^1 f(t) dt$$

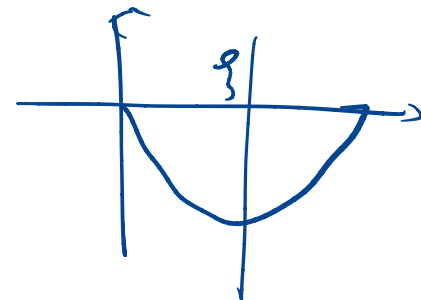
证法一 (利用积分定义) 由于可积, 并注意到 $f(x)$ 单调减, 由此推出

$$= f(x) - f(\xi), \quad \xi \in (0,1)$$

$$f(at) \geq f(a) \quad (0 < a < 1)$$



$$\int_0^a f(x) dx = \lim_{n \rightarrow \infty} \frac{a}{n} \sum_{i=1}^n f\left(\frac{a}{n} i\right) \geq a \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n f\left(\frac{i}{n}\right) = a \int_0^1 f(x) dx.$$



证法二 (利用换元法) 令 $x = at$, 则

$$\int_0^a f(x) dx = a \int_0^1 f(at) dt \geq a \int_0^1 f(t) dt = a \int_0^1 f(x) dx.$$

4. 若 $f(x)$ 在 $[0,1]$ 上连续且单调下降, 证明: 对于任意的 $a \in (0,1)$, 都有

$$\int_0^a f(x)dx \geq a \int_0^1 f(x)dx.$$

证法三 (利用分段积分法) 因 $\int_0^1 f(x)dx = \int_0^a f(x)dx + \int_a^1 f(x)dx$, 故所证不等式等价于

$$\int_0^a f(x)dx - a \int_0^a f(x)dx - a \int_a^1 f(x)dx = (1-a) \int_0^a f(x)dx - a \int_a^1 f(x)dx \geq 0,$$

由积分中值定理:

$$(1-a) \int_0^a f(x)dx - a \int_a^1 f(x)dx = (1-a)a f(\xi) - a(1-a)f(\eta) = (1-a)a[f(\xi) - f(\eta)],$$

因 $0 < \xi < \eta < 1$, 又 $f(x)$ 单调减, 所以上式大于等于零, 即结论成立.

证明: $\int_0^{\frac{\pi}{2}} \sin^m x \cos^m x dx = \frac{1}{2^m} \int_0^{\frac{\pi}{2}} \cos^m x dx$ (m 为正整数).

证: $\because \int_0^{\frac{\pi}{2}} \sin^m x \cos^m x dx = \frac{1}{2^m} \int_0^{\frac{\pi}{2}} \sin^m 2x dx \stackrel{2x=t}{=} \frac{1}{2^{m+1}} \int_0^{\pi} \sin^m t dt$

又证: $\int_0^{\frac{\pi}{2}} \sin^m 2x dx = \int_0^{\frac{\pi}{2}} \cos^m x dx$

令 $2x = \frac{\pi}{2} - u$
 $\Rightarrow u = 2x - \frac{\pi}{2}$
 $\frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^m u du = \int_0^{\frac{\pi}{2}} \cos^m x dx$

$$= \frac{1}{2^{m+1}} \left(\int_0^{\frac{\pi}{2}} \sin^m t dt + \int_{\frac{\pi}{2}}^{\pi} \sin^m t dt \right)$$

$$\int_0^{\frac{\pi}{2}} \sin^m t dt \stackrel{t=\frac{\pi}{2}-u}{=} \int_{\frac{\pi}{2}}^0 \cos^m u (-du) = \int_0^{\frac{\pi}{2}} \cos^m u du = \int_0^{\frac{\pi}{2}} \cos^m x dx,$$

$$\int_{\frac{\pi}{2}}^{\pi} \sin^m t dt \stackrel{t=\pi-u}{=} \int_{\frac{\pi}{2}}^0 \sin^m u (-du) = \int_0^{\frac{\pi}{2}} \sin^m u du = \int_0^{\frac{\pi}{2}} \cos^m x dx,$$

$$\therefore \int_0^{\frac{\pi}{2}} \sin^m x \cos^m x dx = \frac{1}{2^{m+1}} \left(\int_0^{\frac{\pi}{2}} \cos^m x dx + \int_0^{\frac{\pi}{2}} \cos^m x dx \right) = \frac{1}{2^m} \int_0^{\frac{\pi}{2}} \cos^m x dx.$$

定积分不等式的证明

题型二：若仅知被积函数连续：作辅助函数，将结论所含定积分化为变限积分，移项使右边为零，左边即为辅助函数，再用函数单调性或求证。

（微分法）

设 $f(x)$ 是区间 $[a, b]$ 上单调递减的连续函数, 证明:

$$F(x) = \int_a^x (x-a)^3 f(x) dx - \frac{(x-a)^3}{4} \int_a^x f(x) dx$$
$$\int_a^b (x-a)^3 f(x) dx \leq \frac{(b-a)^3}{4} \int_a^b f(x) dx .$$

证: 设 $F(t) = \frac{(t-a)^3}{4} \int_a^t f(x) dx - \int_a^t (x-a)^3 f(x) dx$, $a \leq t \leq b$,

则 $F(0) = 0$,

$$F'(t) = \frac{3}{4}(t-a)^2 \int_a^t f(x) dx + \frac{(t-a)^3}{4} f(t) - (t-a)^3 f(t)$$
$$= \frac{3}{4}(t-a)^2 \left[\int_a^t f(x) dx - (t-a) f(t) \right]$$

由积分中值定理, 得

$$F'(t) = \frac{3}{4}(t-a)^2 (t-a)(f(\xi) - f(t)) , \quad a < \xi < t ,$$

因 $f(x)$ 是区间 $[a, b]$ 上单调递减, 所以 $F'(t) \geq 0$,

于是, $F(b) \geq F(a) = 0$, 即 $\int_a^b (x-a)^3 f(x) dx \leq \frac{(b-a)^3}{4} \int_a^b f(x) dx .$

$$F(x) = (x-a)f\left(\frac{a+x}{2}\right) - \int_a^x f(t)dt$$

若 $f''(x) \geq 0$, $x \in [a, b]$, 证明: $F'(x) = \underbrace{f\left(\frac{a+x}{2}\right)}_{\substack{= [f'(\frac{a+x}{2}) - f'(a)] \frac{x-a}{2} \\ \geq 0}} + \frac{1}{2}(x-a)f'\left(\frac{a+x}{2}\right) - \underbrace{f(x)}_{\geq 0}$

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) dx \leq \frac{f(a) + f(b)}{2}$$

证法 2 由左边不等式, 设 $F(x) = \int_a^x f(t) dt - (x-a)f\left(\frac{a+x}{2}\right)$

$$F'(x) = f(x) - f\left(\frac{a+x}{2}\right) - \frac{x-a}{2} f'\left(\frac{a+x}{2}\right)$$

$$= f'(\xi) \frac{x-a}{2} - \frac{x-a}{2} f'\left(\frac{a+x}{2}\right) \quad \frac{a+x}{2} < \xi < x$$

$$= \frac{x-a}{2} \left[f'(\xi) - f'\left(\frac{a+x}{2}\right) \right] = \frac{x-a}{2} f''(\eta) \left(\xi - \frac{a+x}{2} \right) \geq 0 \quad \frac{a+x}{2} < \eta < \xi$$

故 $F(x)$ 单调不减, $F(b) \geq F(a) = 0 \Rightarrow F(x) \geq 0 \Rightarrow$ 左边得证

由右边不等式, 设 $G(x) = \frac{f(a) + f(x)}{2} (x-a) - \int_a^x f(t) dt$

若 $f''(x) \geq 0$, $x \in [a, b]$, 证明:

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) dx \leq \frac{f(a)+f(b)}{2}$$

$$\begin{aligned} G'(x) &= \frac{1}{2}[(x-a)f'(x) + f(a) - f(x)] \\ &= \frac{1}{2}(x-a)[f'(x) - f'(\xi)] \quad a < \xi < x \\ &= \frac{1}{2}(x-a)(x-\xi)f''(\eta) \geq 0 \quad \xi < \eta < x \end{aligned}$$

故 $G(x)$ 单调不减, $G(b) \geq G(a) = 0 \Rightarrow G(x) \geq 0 \Rightarrow$ 右边得证

综上所述

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) dx \leq \frac{f(a)+f(b)}{2}$$

定积分不等式的证明

题型三：若已知被积函数可导，且至少有一端点 $f(a)=0$

可以用拉格朗日中值定理 $f(x) = f(x) - f(a) = f'(\xi)(x-a)$

或将函数化为变限积分，即 $f(x) = \int_a^x f'(t)dt$;

9. 设 $f'(x)$ 在 $[a, b]$ 上连续, 且 $f(a) = 0$, 证明: $\left| \int_a^b f(x) dx \right| \leq \frac{(b-a)^2}{2} \max |f'(x)|$.

$$\begin{aligned}
 \text{证 因 } \left| \int_a^b f(x) dx \right| &\leq \int_a^b |f(x)| dx = \int_a^b |f(x) - f(a)| dx \\
 &= \int_a^b |f'(\xi)| (x-a) dx \quad (a < \xi < b) \\
 &\leq \max |f'(x)| \int_a^b (x-a) dx \quad (\text{因 } f'(x) \text{ 在 } [a, b] \text{ 上连续}) \\
 &= \frac{(b-a)^2}{2} \max |f'(x)|.
 \end{aligned}$$

$$\begin{aligned}
 \left| \int_a^b f(x) dx \right| &\leq \int_a^b |f(x)| dx \\
 &= \int_a^b |f(x) - f(a)| dx \\
 &= \int_a^b |f'(\xi)| (x-a) dx \\
 &= |f'(\xi)| \left| \frac{1}{2}x^2 - ax \right|_a^b \\
 &= \frac{1}{2} (b-a)^2 |f'(\xi)|
 \end{aligned}$$

5. 设 $f(x)$ 在 $[0,2]$ 上可导, $\underline{\underline{|f'(x)| \leq 1}}$, $f(0) = f(2) = 0$, 证明 $-1 \leq \int_0^2 f(x)dx \leq 1$.

$$\begin{aligned} \text{证 因 } \left| \int_0^2 f(x)dx \right| &\leq \int_0^2 |f(x)| dx = \int_0^1 |f(x) - f(0)| dx + \int_1^2 |f(x) - f(2)| dx \\ &= \int_0^1 |f'(\xi_1)| x dx + \int_1^2 |f'(\xi_2)| (2-x) dx \quad (0 < \xi_1 < 1 < \xi_2 < 2) \leq 1. \end{aligned}$$

$$\begin{aligned} \int_0^2 f(x) dx &= \int_0^1 (f(x) - f(0)) dx + \int_1^2 (f(x) - f(2)) dx = \int_0^1 f'(\xi_1) x dx + \int_1^2 f'(\xi_2) (x-2) dx \\ &= \frac{f'(\xi_1)}{2} - \frac{f'(\xi_2)}{2} \end{aligned}$$

例 29 设 $f(x)$ 在 $[a, b]$ 上连续, 在 (a, b) 内可导, 且 $f'(x) \leq M$, $f(b) = f(a) = 0$,

证明: $\int_a^b |f(x)| dx \leq \frac{M}{4} (b-a)^2$ $\int_a^b |f(x)| dx = \int_a^{\frac{a+b}{2}} |f(x) - f(a)| dx + \int_{\frac{a+b}{2}}^b |f(x) - f(b)| dx$

证法 1 $|f(x)| = |f(x) - f(a)| = \left| \int_a^x f'(t) dt \right| \leq \int_a^x |f'(t)| dt \leq M(x-a)$ $= \int_a^{\frac{a+b}{2}} (x-a) |f'(t)| dt + \int_{\frac{a+b}{2}}^b (b-x) |f'(t)| dt$

$$|f(x)| = |f(x) - f(b)| = \left| \int_x^b f'(t) dt \right| \leq \int_x^b |f'(t)| dt \leq M(b-x)$$

$$\int_a^b |f(x)| dx = \int_a^{\frac{a+b}{2}} |f(x)| dx + \int_{\frac{a+b}{2}}^b |f(x)| dx$$

$$\leq M \int_a^{\frac{a+b}{2}} (x-a) dx + M \int_{\frac{a+b}{2}}^b (b-x) dx = \frac{M}{4} (b-a)^2$$

证法 2 由拉格朗日定理有

$$f(x) = f(x) - f(a) = f'(\xi)(x-a), \quad \xi_1 \in (a, x) \Rightarrow |f(x)| \leq M(x-a)$$

$$f(x) = f(x) - f(b) = f'(\xi)(x-b), \quad \xi_2 \in (x, b) \Rightarrow |f(x)| \leq M(b-x)$$

同上 (略)

例 28 设 $f(x)$, $g(x)$ 在 $[a, b]$ 上连续 且满足关系式 $\int_a^b f(t)dt = \int_a^b g(t)dt$,

$$\int_a^x f(t)dt \geq \int_a^x g(t)dt, \quad x \in [a, b], \quad \text{证明: } \int_a^b xf(x)dx \leq \int_a^b xg(x)dx$$

$F(x) = \int_a^x f(t)dt$
 $G(x) = \int_a^x g(t)dt$

证 设 $F(x) = f(x) - g(x)$, 则 $\int_a^b xF(x)dx = \int_a^b xG(x)dx$

$$G(x) = \int_a^x [f(t) - g(t)]dt = \int_a^x F(t)dt, \quad G'(x) = F(x)$$

$$\text{又 } G(a) = G(b) = 0, \quad G(x) \geq 0, \quad x \in [a, b]$$

$$\begin{aligned} \int_a^b x[f(x) - g(x)]dx &= \int_a^b xF(x)dx = \int_a^b xdG(x) \\ &= xG(x)\Big|_a^b - \int_a^b G(x)dx = -\int_a^b G(x)dx \leq 0 \end{aligned}$$

故

$$\int_a^b xf(x)dx - \int_a^b xg(x)dx \leq 0 \quad \Rightarrow \quad \int_a^b xf(x)dx \leq \int_a^b xg(x)dx$$

或 $\int_a^b [f(x) - g(x)]dx = 0$, $\int_a^x [f(t) - g(t)]dt \geq 0$, 有

$$\int_a^b x[f(x) - g(x)]dx = \int_a^b x d \int_a^x [f(t) - g(t)]dt$$

$$= x \int_a^x [f(t) - g(t)]dt \Big|_a^b - \int_a^b \left(\int_a^x (f(t) - g(t))dt \right) dx$$

$$= - \int_a^b \left(\int_a^x (f(t) - g(t))dt \right) dx \leq 0$$

定积分不等式的证明

题型四：若已知被积函数二阶可导：将被积函数按泰勒公式展开并放缩，利用定积分比较定理求证。

60. 设 $f(x)$ 是区间 $[0,1]$ 上二阶可导, 且 $f(0) = f(1) = 0$, $|f''(x)| \leq M$ (M 为一个正常数).

证: $\left| \int_0^1 f(x) dx \right| \leq \frac{M}{12}.$

$\forall x \in (0,1)$, 由泰勒公式得:

$$f(0) = f(x) + f'(x)(-x) + \frac{f''(\xi)}{2} x^2, \quad \xi \text{ 介于 } 0 \text{ 与 } x \text{ 之间.}$$

两边在区间 $[0,1]$ 上积分, 有 $0 = \int_0^1 f(x) dx - \int_0^1 f'(x)x dx + \frac{1}{2} \int_0^1 f''(\xi)x^2 dx,$

用分部积分得到 $\int_0^1 f'(x)x dx = x f(x) \Big|_0^1 - \int_0^1 f(x) dx = - \int_0^1 f(x) dx,$

故有 $\int_0^1 f(x) dx = - \frac{1}{4} \int_0^1 f''(\xi)x^2 dx,$

所以 $\left| \int_0^1 f(x) dx \right| \leq \frac{1}{4} \int_0^1 |f''(\xi)|x^2 dx \leq \frac{M}{4} \int_0^1 x^2 dx = \frac{M}{12}.$

64. 已知 $f''(x) \leq 0$, n 为自然数, 证明: $\int_0^1 f(x^n) dx \leq f(\frac{1}{n+1})$.

证 记 $t_0 = \frac{1}{n+1}$, 对任给 $t \in [0,1]$, 由台劳公式, $\exists \eta \in (0,1)$, 使

$$f(t) = f(t_0) + f'(t_0)(t - t_0) + \frac{1}{2} f''(\eta)(t - t_0)^2 \text{ 由 } f''(x) \leq 0, \text{ 便有}$$

$f(t) \leq f(t_0) + f'(t_0)(t - t_0)$ (由凸曲线在切线之下性质也可直达此处) 于是, 取

$t = x^n, 0 \leq x \leq 1$, 得 $f(x^n) \leq f(t_0) + f'(t_0)(x^n - t_0)$, 以及

$$\int_0^1 f(x^n) dx \leq \int_0^1 (f(t_0) + f'(t_0)(x^n - t_0)) dx = f(t_0).$$

68. 设 $f(x)$ 在区间 $[0, 2]$ 上的二阶导函数连续, $f(0) = f(2) = 0$ 。

证明: $\left| \int_0^2 f(x) dx \right| \leq \frac{2M}{3}$, 其中 $M = \max_{0 \leq x \leq 2} |f''(x)|$ 。

证 选择在任意点 $x \in [0, 2]$ 处展开:

$$f(t) = f(x) + f'(x)(t-x) + \frac{f''(\xi)}{2}(t-x)^2, \quad t \in [0, 2]$$

令 $t=0$ 代入, 得 $0 = f(x) - f'(x)x + \frac{f''(\xi)}{2}x^2$

两边积分 $\int_0^2 f(x) dx = \int_0^2 f'(x)x dx - \frac{1}{2} \int_0^2 f''(\xi)x^2 dx$

其中 $\int_0^2 f'(x)x dx = \int_0^2 x df(x) = xf(x)|_0^2 - \int_0^2 f(x) dx = -\int_0^2 f(x) dx$

故 $\int_0^2 f(x) dx = -\frac{1}{4} \int_0^2 f''(\xi)x^2 dx$

从而取绝对值后放大积分, 得证

$$\left| \int_0^2 f(x) dx \right| \leq \frac{1}{4} M \int_0^2 x^2 dx = \frac{2}{3} M$$

定积分不等式的证明

题型五： 含有平方的不等式可以考虑柯西不等式。

$$; \left[\int_a^b f(x)g(x) dx \right]^2 \leq \int_a^b f^2(x) dx \int_a^b g^2(x) dx$$

(1) 设 $f(x)$ 在 $[a, b]$ 上有连续导数, 且 $f(a) = 0$, 证明:

$$\int_a^b f^2(x) dx \leq \frac{(b-a)^2}{2} \int_a^b \underbrace{[f'(x)]^2}_{\text{blue underline}} dx$$

证 $f(x) = f(x) - f(a) = \int_a^x f'(t) dt, \quad x \in [a, b]$

$$f^2(x) = \left[\int_a^x f'(t) dt \right]^2 \leq \int_a^x 1^2 dt \int_a^x [f'(t)]^2 dt$$

$$= (x-a) \int_a^x [f'(t)]^2 dt \leq (x-a) \int_a^b [f'(x)]^2 dx$$

$$\int_a^b f^2(x) dx \leq \int_a^b (x-a) dx \int_a^b [f'(x)]^2 dx = \frac{(b-a)^2}{2} \int_a^b [f'(x)]^2 dx$$

(3) 设 $f(x)$ 在 $[a, b]$ 上连续, 且 $f(x) \geq 0$, $\int_a^b f(x)dx = 1$, 证明

$$\left(\int_a^b f(x) \sin \lambda x dx\right)^2 + \left(\int_a^b f(x) \cos \lambda x dx\right)^2 \leq 1$$

证 $\left[\int_a^b f(x) \sin \lambda x dx\right]^2 = \left[\int_a^b \sqrt{f(x)} \sqrt{f(x)} \sin \lambda x dx\right]^2$

$$\leq \int_a^b f(x) dx \int_a^b f(x) \sin^2 \lambda x dx = \int_a^b f(x) \sin^2 \lambda x dx$$

同理有 $\left[\int_a^b f(x) \sin \lambda x dx\right]^2 \leq \int_a^b f(x) \cos^2 \lambda x dx$

$$\begin{aligned} & \left[\int_a^b f(x) \sin \lambda x dx\right]^2 + \left[\int_a^b f(x) \cos \lambda x dx\right]^2 \\ & \leq \int_a^b f(x) \sin^2 \lambda x dx + \int_a^b f(x) \cos^2 \lambda x dx \\ & = \int_a^b f(x) (\sin^2 \lambda x + \cos^2 \lambda x) dx = \int_a^b f(x) dx = 1 \end{aligned}$$

$$g'(x) = -\frac{1}{(x-a)^2} \int_a^x f(t) dt + \frac{1}{x-a} f(x) = \frac{1}{x-a} [f(x) - f(\xi)]$$

6. 设 $f(x)$ 在 $[a, b]$ 上连续且单调增加, 证明: $g(x) = \frac{1}{x-a} \int_a^x f(t) dt$ 在 $[a, b]$ 上单调增加.

$$\text{证 } g'(x) = \frac{f(x)(x-a) - \int_a^x f(t) dt}{(x-a)^2} = \frac{f(x)(x-a) - f(\xi)(x-a)}{(x-a)^2} = \frac{f(x) - f(\xi)}{x-a},$$

其中 $a < \xi < x$. 因 $f(x)$ 在 $[a, b]$ 上单调增加, 故 $f(x) - f(\xi) \geq 0$,

从而 $g'(x) \geq 0$, 即 $g(x)$ 在 $[a, b]$ 上单调增加.

5. 设函数 $f(x)$ 在 $[A, B]$ 上连续, 证明 $\lim_{h \rightarrow 0} \frac{1}{h} \int_a^x [f(t+h) - f(t)] dt = f(x) - f(a)$,

$$A < a < x < B.$$

证 因 $f(x)$ 在 $[A, B]$ 上连续, 故存在原函数 $F(x)$, 因此

$$\int_a^x [f(t+h) - f(t)] dt = [F(t+h) - F(t)]_a^x$$

于是, 原极限 $= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x) - [F(a+h) - F(a)]}{h} = F'(x) - F'(a) = f(x) - f(a).$

6. 设函数 $f(x)$ 连续, 证明 $\int_0^x (x-u)f(u)du = \int_0^x [\int_0^u f(t)dt] du.$

$$\begin{aligned} \text{证 右} &= u \int_0^u f(t) dt \Big|_0^x - \int_0^x u f(u) du \\ &= x \int_0^x f(u) du - \int_0^x u f(u) du \\ &= \int_0^x (x-u)f(u) du = \text{左}. \end{aligned}$$

$x \int_0^x f(u) du - \int_0^x u f(u) du$
 $\int_0^x f(u) du \quad \text{---} \quad \int_0^x f(u) du$
 $\Rightarrow$

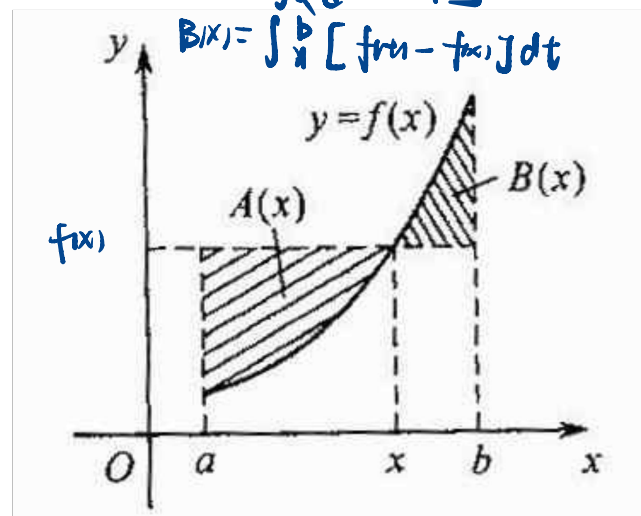
7. 设 $f(x)$ 在 $[a, b]$ 上可导, 且 $f'(x) > 0$, $f(a) > 0$. 证明: 对如图所示的两个面积函数 $A(x)$

$f(x) \cdot t -$

$$A(x) = \int_a^x [f(x) - f(t)] dt$$

$$B(x) = \int_x^b [f(t) - f(x)] dt$$

与 $B(x)$ ($a \leq x \leq b$), 存在唯一的 $\xi \in (a, b)$, 使得 $\frac{A(\xi)}{B(\xi)} = 2017$.



解 由 $f(x)$ 在 $[a, b]$ 上可导, $f'(x) > 0$, $f(a) > 0$ 得 $f(x)$

在 $[a, b]$ 上单调增加, 且 $f(x) > 0$. 依题意:

$$A(x) = f(x)(x-a) - \int_a^x f(t) dt; \quad B(x) = \int_x^b f(t) dt - f(x)(b-x).$$

设 $F(x) = A(x) - 2017B(x)$, 则 $F(x)$ 在 $[a, b]$ 上连续。因

$$F'(x) = f'(x)(x-a) + f(x) - f(x) - 2017[-f(x) - f'(x)(b-x) + f(x)],$$

$= f'(x)(x-a) + 2017 f'(x)(b-x) > 0$, 即 $F(x)$ 在 $[a, b]$ 单调增加。

又 $F(a) = -2017 \int_a^b [f(t) - f(a)] dt < 0$; $F(b) = \int_a^b [f(b) - f(t)] dt > 0$,

由零点定理, 存在唯一 $\xi \in (a, b)$, 使得 $F(\xi) = 0$, 即 $A(\xi) = 2017B(\xi)$.